Data Analysis for Public Policy
Group Report #1

Operationalisation, Descriptives, and Comparing Two Groups

For the first group report, you will operationalise the research question your group signed up for in class, describe your data, build a grouping variable, and run a first comparison between two groups using a t-test.
You will hand in the report **one week before the midterm exam - October 28**, together with the data and the R code you used. The report is graded and counts as 5% of your final course grade.

Why this report matters. During the midterm exam you will work directly with your own report. We will give you the report and ask you to extend or reinterpret parts of it. The goal is therefore not to produce something that merely looks finished, but to set up variables and code that you understand and feel comfortable interpreting.
A suggested best practice is to choose items you can explain, that fit the required criteria, and make sure you know what your code does, and the steps necessary to produce outputs in your report.

1. Operationalisation: turning your research question into measurable variables

Take the constructs in your research question and evaluate them using concrete survey items from the European Social Survey. You must use data from either **ESS Round 10 or Round 11 for Switzerland**. Both datasets and their codebooks are available on Moodle. All of the research questions are answerable with items from either, or both, of these data. But in your analyses, you can only use **ONE** of these datasets, not both.

Search the codebook and identify **two individual survey items**.

1. **Dependent variable (DV):** the outcome from your research question that you want to explain. This must be a continuous item or a Likert-type item.

2. **Independent variable (IV):** the main factor from your research question that you think is causing variation in the outcome. This must also be a continuous item or a Likert-type item.

For **each** variable, explain in a few short sentences:

- why you operationalised the construct in this way;
- why you think the selected item captures what your research question is asking about; and
- at least one potential limitation of this operationalisation.

No measure is perfect, especially when using secondary survey data that you did not design yourself. The question may not capture the construct exactly as you would like, the response categories may be coarse, or the wording may introduce ambiguity.

Example
Research Question: Are people who are more satisfied with the state of the economy also more satisfied with life overall?

- **DV:** Life satisfaction, 0–10 (`stflife`)
- **IV:** Satisfaction with the economy, 0–10 (`stfeco`)

Always confirm the exact variable name, question wording, and coding against the codebook for the ESS round you are using.

2. Creating your grouping variable

A t-test compares **two groups**, which is inherently a comparison in how a continuous dependent variable differs by a binary (2-group) categorical variable. But your independent variable is continuous (which we will need for the second half of the course). Therefore, you will create a new binary (`0/1`) grouping variable out of your independent variable by splitting respondents into:

- respondents **above the mean** of the independent variable; and
- respondents **at or below the mean**.

In your R script, keep the original continuous variable unchanged and create the binary version as a **new variable**. This should make changing the cut-point a one-line modification that you can re-run and is replicable. In the report, clearly state how you handled this transformation—for example if some observations are exactly equal to the mean.

3. Descriptive statistics

Describe the following three variables:

1. your dependent variable;
2. your original, continuous independent variable; and
3. the binary grouping variable you created.

Report statistics appropriate to each variable's level of measurement.

Variable	What to report
Dependent variable	Valid N; mean; median; mode; variance; standard deviation; range; IQR; histogram.
Continuous independent variable	Valid N; mean; median; mode; variance; standard deviation; range; IQR; histogram.
Binary grouping variable	Valid N; count and proportion in each group; mode; bar chart.

Note: for the binary grouping variable, **do not report** the mean, standard deviation, variance, range, or IQR. These are not substantively meaningful for a categorical variable.

Make sure that you produce **every plot using `ggplot`**.

Then, in one or two sentences in the report, read the descriptives back to us. Mention any recoding you performed, whether the split produced two reasonably sized groups, and anything notable in the data—for example unusual distributions, obvious outliers, response heaping, or substantial missingness.

4. Comparing two groups: the t-test

Use a t-test to compare your dependent variable across the two groups defined by the binary grouping variable.

Your substantive question is:

> *Is the average level of the dependent variable different for respondents on one side of the binary variable compared with respondents on the other side of the binary variable?*

You must decide which t-test is appropriate and explain why.

5. Interpreting your result

Interpret the t-test in prose. Your discussion should cover:

1. which two groups you are comparing and the mean of the dependent variable in each;
2. the difference between the group means, its direction, the test statistic, and the p-value;
3. what the result means substantively for your research question, in plain language rather than only "significant" or "not significant"; and
4. the limitation of the cut-point.

Because you split a continuous independent variable at its mean, the t-test uses only a crude two-group version of the variable and ignores all variation within the two groups. Explain what this means for how much weight the result can bear, and note that regression in Report 2 will use the independent variable in full.

6. What you need to hand in

1. **Written report.** No more than **one page of text**, or **600 words**. Plots, tables, and code do *not* count towards this limit.
   The report should include:
   - a project title;
   - the names of all group members;
   - your research question;
   - the operationalisation of the two variables, including justification and limitations;
   - a short account of how you created the grouping variable;
   - descriptive statistics and plots, with a short interpretation;
   - the t-test; and
   - your substantive interpretation of the result.
2. **R script.** The script must run from the raw data provided (`ESS10_CHData.csv` or `ESS11_CHData.csv`) all the way through every recode, statistic, and plot used in the report. It must be fully reproducible from start to finish. Comment the code so that we can follow what each step does.

Write the report in **full sentences, not bullet points**. The title and list of group members are the only exceptions. Be concise: a clear, short report is better than a long one.

7. Using R and AI tools

If you want, you may use AI coding as an assisting tool as you are writing this report. However, you are responsible for **every line of your script** and **every claim in your report**, and you must be able to explain both.

Remember that the midterm exam is built on this report. If you cannot explain what your code does or why you selected a particular variable, the exam will be much harder.

Use these tools to help you understand how the analysis works, not to skip that understanding.

8. Submission and grading

Weight: 5% of your final grade
Due date: 28 October 2026
Submit by email to:
keith.smith@gess.ethz.ch
cfournier@ethz.ch

9. Grading rubric

The report is marked out of **20 points**. You will be graded upon how well the report conducts each step correctly: selecting a valid variable type, reporting the right statistic for each variable, and running the appropriate test.

Component	What earns the marks	Marks
Operationalisation	The DV and IV are each an appropriate continuous or Likert-type item (2); a clear justification for each item (2); at least one genuine limitation noted per variable (1).	5
Grouping variable	Binary variable correctly created by splitting the IV at its mean in R (2); the original continuous variable is left intact and the boundary handling is stated (1).	3
Descriptive statistics	Valid N for all three variables (1); correct central tendency and spread for the DV and continuous IV (2); correct descriptives for the binary variable, with no meaningless statistics reported (1); the appropriate `ggplot` for each variable (1).	5
T-test	An appropriate t-test is correctly run in R (2); the choice of test is justified (1).	3
Interpretation	The result is reported in full—groups, means, difference, direction, test statistic, p-value (1); a substantive reading plus the cut-point limitation (1).	2
Code and reproducibility	The script runs from the raw data through to every reported result (1); the code is commented and readable (1).	2
Total		**20**

The analysis marks depend on your code. Every statistic, plot, and test above is credited only if your script actually produces it. A number that appears in the report but is not generated by the code, or a script that does not run, cannot receive the marks. Keeping the code you share with us short also makes sure we can easily identify the relevant information.

If you have any questions, or if anything is unclear, please contact Camille and Keith.